

DECENT COACHING CLASSES

MATHS BASIC FOUNDATION NOTES

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1. Number Systems

1.1 Classification of Numbers

- Natural numbers (N):** 1, 2, 3, 4, ... (counting numbers)
- Whole numbers (W):** 0, 1, 2, 3, ... (natural numbers + zero)
- Integers (Z):** ..., -3, -2, -1, 0, 1, 2, 3, ... (whole numbers + negative numbers)
- Rational numbers (Q):** numbers that can be written as p/q , where p, q are integers and $q \neq 0$
- Irrational numbers:** numbers that cannot be written as p/q , e.g. $\sqrt{2}$, $\sqrt{3}$, π
- Real numbers (R):** all rational and irrational numbers together

1.2 Place Value and Face Value

- Face value** of a digit is the digit itself.
- Place value** of a digit = face value $\times$ value of its position (ones, tens, hundreds, thousands, ten-thousands, lakhs, crores, or ones/tens/hundreds/thousand/million in the international system).

Note: Indian system: Ones, Tens, Hundreds, Thousands, Ten Thousands, Lakhs, Ten Lakhs, Crores. International system: Ones, Tens, Hundreds, Thousands, Ten Thousands, Hundred Thousands, Millions.

1.3 Types of Whole Numbers

- Even numbers:** divisible by 2 (0, 2, 4, 6, ...)
- Odd numbers:** not divisible by 2 (1, 3, 5, 7, ...)
- Prime numbers:** exactly two factors, 1 and itself (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ...). 2 is the only even prime number.
- Composite numbers:** more than two factors (4, 6, 8, 9, 10, ...). 1 is neither prime nor composite.
- Co-prime (relatively prime) numbers:** two numbers whose HCF is 1, e.g. 8 and 15
- Twin primes:** pairs of prime numbers differing by 2, e.g. (3,5), (5,7), (11,13)

1.4 Divisibility Rules

Divisible by	Rule
2	Last digit is 0, 2, 4, 6 or 8
3	Sum of digits is divisible by 3
4	Number formed by last 2 digits is divisible by 4
5	Last digit is 0 or 5
6	Divisible by both 2 and 3
8	Number formed by last 3 digits is divisible by 8
9	Sum of digits is divisible by 9
10	Last digit is 0
11	Difference of (sum of digits at odd places) and (sum of digits at even places) is 0 or divisible by 11

1.5 Factors, Multiples, HCF and LCM

- Factor:** a number that divides another number exactly (no remainder)
- Multiple:** a number obtained by multiplying a given number by 1, 2, 3, ...
- HCF (Highest Common Factor)** — also called GCD — is the greatest number that divides two or more numbers exactly. Found by prime factorisation (product of common prime factors with lowest powers) or long division method.
- LCM (Lowest Common Multiple)** is the smallest number that is a multiple of two or more numbers. Found by prime factorisation (product of all prime factors with highest powers) or common division method.
- Important relation:** for two numbers a and b — $HCF(a, b) \times LCM(a, b) = a \times b$

1.6 Integers — Rules of Operations

- Addition:** same signs $\rightarrow$ add and keep the sign; different signs $\rightarrow$ subtract and take the sign of the bigger number

- **Subtraction:** change the sign of the number being subtracted and add, e.g. $a - b = a + (-b)$
- **Multiplication/Division:** same signs $\rightarrow$ answer is positive; different signs $\rightarrow$ answer is negative ($++ = +$, $-- = +$, $+- = -$, $-+ = -$)

1.7 Order of Operations — BODMAS

- Brackets $\rightarrow$ Of $\rightarrow$ Division $\rightarrow$ Multiplication $\rightarrow$ Addition $\rightarrow$ Subtraction (solve in this order)

2. Fractions and Decimals

2.1 Types of Fractions

- **Proper fraction:** numerator $<$ denominator, e.g. $3/5$
- **Improper fraction:** numerator $\geq$ denominator, e.g. $7/4$
- **Mixed fraction:** whole number + proper fraction, e.g. $1\frac{3}{4}$
- **Like fractions:** same denominator, e.g. $2/7$, $5/7$
- **Unlike fractions:** different denominators, e.g. $2/3$, $3/5$
- **Equivalent fractions:** different fractions representing the same value, e.g. $1/2 = 2/4 = 3/6$

2.2 Operations on Fractions

- **Addition/Subtraction:** convert to like fractions (same denominator using LCM) then add/subtract numerators
- **Multiplication:** multiply numerators together and denominators together — $(a/b) \times (c/d) = ac/bd$
- **Division:** multiply by the reciprocal of the second fraction — $(a/b) \div (c/d) = (a/b) \times (d/c)$
- **Reciprocal:** of a/b is b/a (interchange numerator and denominator)

2.3 Decimal Numbers

- Place value chart after the decimal point: **tenths ($1/10$)**, **hundredths ($1/100$)**, **thousandths ($1/1000$)**
- **Fraction to decimal:** divide numerator by denominator
- **Decimal to fraction:** write digits after the decimal point over 10, 100, 1000 (as per number of decimal places) and simplify
- **Terminating decimal:** decimal expansion ends, e.g. 0.5, 0.25
- **Repeating (recurring) decimal:** a digit or group of digits repeats endlessly, e.g. 0.333...

2.4 Operations on Decimals

- **Addition/Subtraction:** align the decimal points, then add or subtract as usual
- **Multiplication:** multiply ignoring the decimal point, then place the decimal point in the answer by counting total decimal places in both numbers
- **Division:** convert the divisor to a whole number by shifting the decimal point in both divisor and dividend, then divide

3. Ratio, Proportion, Percentage and Unitary Method

3.1 Ratio

- Ratio compares two quantities of the **same kind and same units** — $a : b = a/b$
- A ratio is simplified by dividing both terms by their HCF
- Ratios have no units

3.2 Proportion

- Four quantities a , b , c , d are in proportion if $a : b = c : d$, written $a : b :: c : d$
- **Product of extremes = Product of means:** $a \times d = b \times c$
- **Continued proportion:** a , b , c are in continued proportion if $a : b = b : c$, so $b^2 = a \times c$ (b is the mean proportional)

3.3 Percentage

- Percentage means "per hundred" (denoted %)
- **Fraction to percentage:** multiply the fraction by 100

- **Percentage to fraction:** divide by 100 and simplify
- **Decimal to percentage:** multiply by 100
- To find x% of a quantity Q: $(x/100) \times Q$

3.4 Unitary Method

- Used to find the value of a single unit first, then the value of the required number of units. Steps: find the value of ONE unit (divide), then multiply by the required number of units.

4. Exponents and Powers

4.1 Meaning

- For a number a raised to power n (a^n), **a is the base** and **n is the exponent (or power/index)**. a^n means a multiplied by itself n times.

4.2 Laws of Exponents

Law	Rule
Multiplying powers with same base	$a^m \times a^n = a^{(m+n)}$
Dividing powers with same base	$a^m \div a^n = a^{(m-n)}$
Power of a power	$(a^m)^n = a^{(mn)}$
Power of a product	$(ab)^n = a^n \times b^n$
Power of a quotient	$(a/b)^n = a^n / b^n$
Zero exponent	$a^0 = 1$ ($a \neq 0$)
Negative exponent	$a^{-n} = 1 / a^n$
Power one	$a^1 = a$

4.3 Standard Form (Scientific Notation)

- A large or small number written as $k \times 10^n$, where $1 \leq k < 10$ and n is an integer, e.g. $149,600,000 = 1.496 \times 10^8$

5. Squares, Square Roots, Cubes and Cube Roots

5.1 Perfect Squares (1^2 to 20^2)

n	n^2	n	n^2
1	1	11	121
2	4	12	144
3	9	13	169
4	16	14	196
5	25	15	225
6	36	16	256
7	49	17	289
8	64	18	324
9	81	19	361
10	100	20	400

5.2 Properties of Square Numbers

- A number ending in 2, 3, 7 or 8 is never a perfect square
- A perfect square always ends in 0, 1, 4, 5, 6 or 9
- The square of an even number is even; the square of an odd number is odd
- Between n^2 and $(n+1)^2$ there are $2n$ non-perfect-square numbers

5.3 Finding Square Roots

- **Prime factorisation method:** express the number as a product of prime factors, pair identical factors, and take one from each pair
- **Long division method:** used for larger numbers or non-perfect squares (to get decimal approximations)
- $\sqrt{\quad}$ is the symbol for square root; square root is the inverse operation of squaring

5.4 Perfect Cubes (1^3 to 10^3)

n	n^3
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512
9	729
10	1000

5.5 Cube Roots

- **Cube root ($\sqrt[3]{\quad}$)** is the inverse of cubing. Found by prime factorisation — group identical prime factors in triples and take one from each triple.

Note: A number ending in 0 has a cube ending in 0; ending in 1 $\rightarrow$ cube ends in 1; ending in 4 $\rightarrow$ cube ends in 4, and so on — useful for estimating cube roots of perfect cubes.

6. Algebra — Basic Concepts

6.1 Key Terms

- **Constant:** a fixed numerical value, e.g. 5, -3, 7
- **Variable:** a symbol (usually a letter like x, y, z) that can take different values
- **Algebraic expression:** a combination of constants and variables connected by +, -, $\times$, $\div$, e.g. $3x + 5y - 7$
- **Term:** parts of an expression separated by + or - signs, e.g. in $3x + 5y$, the terms are $3x$ and $5y$
- **Factors of a term:** numbers/variables multiplied to form the term, e.g. $5xy$ has factors 5, x, y
- **Coefficient:** the numerical factor of a term, e.g. in $7xy$, the coefficient of xy is 7
- **Like terms:** terms having the same variables with the same powers, e.g. $3x^2$ and $-5x^2$
- **Unlike terms:** terms with different variables or powers, e.g. $3x$ and $3x^2$

6.2 Types of Algebraic Expressions (by number of terms)

- **Monomial:** one term, e.g. $5x$
- **Binomial:** two terms, e.g. $3x + 2y$
- **Trinomial:** three terms, e.g. $x^2 + 2x + 1$
- **Polynomial:** general term for expressions with one or more terms and non-negative integer powers

6.3 Operations on Algebraic Expressions

- **Addition/Subtraction:** only like terms can be added or subtracted; add/subtract their coefficients
- **Multiplication:** monomial $\times$ monomial — multiply coefficients and add powers of same variables using laws of exponents; use distributive law for binomials/polynomials
- **Division:** divide term by term, cancelling common factors

7. Algebraic Identities

An identity is an equation that is true for all values of the variables involved. These standard identities are essential building blocks for Class 9 algebra.

Identity	Expansion
$(a + b)^2$	$a^2 + 2ab + b^2$
$(a - b)^2$	$a^2 - 2ab + b^2$
$(a + b)(a - b)$	$a^2 - b^2$
$(x + a)(x + b)$	$x^2 + (a + b)x + ab$
$(a + b)^3$	$a^3 + 3a^2b + 3ab^2 + b^3$
$(a - b)^3$	$a^3 - 3a^2b + 3ab^2 - b^3$

Note: These identities allow quick multiplication/squaring of numbers close to round values, e.g. $102^2 = (100+2)^2 = 100^2 + 2 \times 100 \times 2 + 2^2 = 10404$, and are the foundation for Class 9's Ch. 2 (Polynomials).

8. Linear Equations in One Variable

- An **equation** has an "=" sign and states two expressions are equal; an **expression** has no "=" sign
- A **linear equation in one variable** has the variable raised to the power 1 only, e.g. $2x + 3 = 11$
- Rule:** whatever operation is done on one side of an equation must be done on the other side, to keep it balanced
- Transposition method:** a term can be moved from one side to the other by changing its sign (+ becomes -, $\times$ becomes $\div$, and vice versa)
- Steps to solve:** (1) simplify both sides, (2) collect variable terms on one side and constants on the other using transposition, (3) simplify to find the value of the variable

Note: Always verify the solution by substituting it back into the original equation.

9. Basic Geometrical Concepts — Lines and Angles

9.1 Basic Terms

- Point:** an exact location, has no length, breadth or thickness
- Line segment:** a straight path between two fixed points (has a definite length)
- Line:** a line segment extended endlessly in both directions
- Ray:** a straight path that starts at a point and extends endlessly in one direction
- Plane:** a flat surface that extends endlessly in all directions

9.2 Types of Angles

Angle	Measure
Acute angle	between 0° and 90°
Right angle	exactly 90°
Obtuse angle	between 90° and 180°
Straight angle	exactly 180°
Reflex angle	between 180° and 360°
Complete angle	exactly 360°

9.3 Pairs of Angles

- Complementary angles:** two angles whose sum is 90°
- Supplementary angles:** two angles whose sum is 180°
- Adjacent angles:** angles with a common vertex and a common arm, lying on opposite sides of the common arm
- Linear pair:** adjacent angles whose non-common arms form a straight line (sum = 180°)
- Vertically opposite angles:** formed when two lines intersect; they are always equal

9.4 Parallel Lines Cut by a Transversal

- **Corresponding angles** are equal
- **Alternate interior angles** are equal
- **Alternate exterior angles** are equal
- **Co-interior (allied) angles** (on the same side of the transversal) are supplementary — sum to 180°

10. Triangles

10.1 Classification by Sides

- **Scalene triangle:** all sides of different lengths
- **Isosceles triangle:** two sides equal (angles opposite equal sides are also equal)
- **Equilateral triangle:** all three sides equal (all angles = 60°)

10.2 Classification by Angles

- **Acute-angled triangle:** all angles less than 90°
- **Right-angled triangle:** one angle exactly 90°
- **Obtuse-angled triangle:** one angle greater than 90°

10.3 Important Properties

- **Angle Sum Property:** the sum of all three interior angles of a triangle is always 180°
- **Exterior Angle Property:** an exterior angle of a triangle equals the sum of the two interior opposite angles
- **Median:** a line segment joining a vertex to the midpoint of the opposite side
- **Altitude:** a perpendicular line segment from a vertex to the opposite side (or its extension)
- **Sum of any two sides** of a triangle is always greater than the third side (triangle inequality)

10.4 Congruence of Triangles

- Two triangles are congruent if they are exactly equal in shape and size. Criteria:
 - **SSS (Side-Side-Side):** all three sides equal
 - **SAS (Side-Angle-Side):** two sides and the included angle equal
 - **ASA (Angle-Side-Angle):** two angles and the included side equal
 - **RHS (Right angle-Hypotenuse-Side):** right angle, hypotenuse and one side equal (right triangles only)

10.5 Pythagoras Theorem

- In a right-angled triangle: **(Hypotenuse)² = (Base)² + (Height)²** i.e. $c^2 = a^2 + b^2$.

11. Quadrilaterals and Polygons

11.1 Polygons

- A **polygon** is a closed figure made of line segments
- Classified by number of sides: Triangle (3), Quadrilateral (4), Pentagon (5), Hexagon (6), Heptagon (7), Octagon (8), Nonagon (9), Decagon (10)
- **Sum of interior angles of a polygon with n sides** = $(n - 2) \times 180^\circ$
- **Regular polygon:** all sides and all angles equal

11.2 Quadrilaterals — Properties

- **Sum of all interior angles of a quadrilateral** = 360°

Shape	Key Properties
Parallelogram	Opposite sides equal & parallel; opposite angles equal; diagonals bisect each other
Rectangle	Parallelogram with all angles 90° ; diagonals equal and bisect each other
Square	Rectangle with all sides equal; diagonals equal, bisect each other at 90°

Shape	Key Properties
Rhombus	Parallelogram with all sides equal; diagonals bisect each other at 90°
Trapezium	One pair of opposite sides parallel
Kite	Two pairs of adjacent sides equal; diagonals intersect at 90°

12. Mensuration

12.1 Perimeter and Area — Plane Figures

Shape	Perimeter	Area
Square (side a)	$4a$	a^2
Rectangle (l, b)	$2(l + b)$	$l \times b$
Triangle (sides a,b,c; base b, height h)	$a + b + c$	$\frac{1}{2} \times \text{base} \times \text{height}$
Equilateral triangle (side a)	$3a$	$(\sqrt{3}/4) a^2$
Parallelogram (base b, height h)	$2(a + b)$	$\text{base} \times \text{height}$
Rhombus (diagonals d1, d2)	$4 \times \text{side}$	$\frac{1}{2} \times d1 \times d2$
Trapezium (parallel sides a, b; height h)	sum of all sides	$\frac{1}{2} \times (a + b) \times h$
Circle (radius r)	$2\pi r$ (circumference)	πr^2

Note: Take $\pi \approx 22/7$ or 3.14 as specified. Diameter = $2 \times \text{radius}$.

12.2 Surface Area and Volume — Solid Shapes

Shape	Surface Area	Volume
Cube (side a)	Total: $6a^2$ Lateral: $4a^2$	a^3
Cuboid (l, b, h)	Total: $2(lb + bh + hl)$ Lateral: $2h(l + b)$	$l \times b \times h$
Cylinder (radius r, height h)	Total: $2\pi r(r + h)$ Curved: $2\pi rh$	$\pi r^2 h$

13. Data Handling — Statistics and Probability

13.1 Basic Terms

- Data:** a collection of numbers/facts gathered to give information
- Tally marks:** used to count data in groups of 5 (I, II, III, IIII, IIII+)
- Frequency:** the number of times a particular observation occurs
- Range:** difference between the highest and lowest values in the data

13.2 Graphical Representation

- Pictograph:** data represented using pictures/symbols
- Bar graph:** data represented using bars of uniform width with equal spacing; height/length proportional to the value
- Double bar graph:** compares two sets of data using paired bars
- Pie chart:** data represented as sectors of a circle, showing parts of a whole
- Histogram:** bar graph for grouped/continuous data with no gaps between bars

13.3 Measures of Central Tendency

- Mean (Average) = Sum of all observations $\div$ Number of observations**
- Median:** the middle value when data is arranged in ascending/descending order (average of two middle values if the number of observations is even)
- Mode:** the observation that occurs most frequently in the data

13.4 Probability (Introductory)

- **Probability of an event** = $\text{Number of favourable outcomes} \div \text{Total number of possible outcomes}$. Probability always lies between 0 and 1.

14. Commercial Mathematics

14.1 Profit and Loss

Term	Formula
Cost Price (CP)	the price at which an article is purchased
Selling Price (SP)	the price at which an article is sold
Profit	$SP - CP$ (when $SP > CP$)
Loss	$CP - SP$ (when $CP > SP$)
Profit %	$(\text{Profit} / CP) \times 100$
Loss %	$(\text{Loss} / CP) \times 100$

14.2 Discount

- **Marked Price (MP)**: the price displayed/listed on an article (before discount)
- **Discount** = $MP - SP$
- **Discount %** = $(\text{Discount} / MP) \times 100$

14.3 Simple Interest and Compound Interest

Term	Formula
Simple Interest (SI)	$SI = (P \times R \times T) / 100$
Amount (Simple Interest)	$A = P + SI$
Compound Interest — Amount	$A = P \times (1 + R/100)^n$
Compound Interest	$CI = A - P$

Note: P = Principal, R = Rate of interest per annum, T = Time in years, n = number of years/conversion periods.

15. Symmetry and Visualising Solid Shapes

15.1 Symmetry

- **Line of symmetry (reflection symmetry)**: a line that divides a figure into two identical mirror-image halves
- **Rotational symmetry**: a figure that looks the same after being rotated by less than 360° about a fixed point (the centre)
- **Order of rotational symmetry**: the number of times a figure matches itself in one full rotation (360°)

15.2 3-D Shapes — Faces, Edges, Vertices

Solid	Faces (F)	Edges (E)	Vertices (V)
Cube	6	12	8
Cuboid	6	12	8
Cylinder	3 (2 flat + 1 curved)	2	0
Cone	2 (1 flat + 1 curved)	1	1
Sphere	1 (curved)	0	0
Triangular Pyramid (Tetrahedron)	4	6	4
Square Pyramid	5	8	5

15.3 Euler's Formula

- For any polyhedron: $F + V - E = 2$ (Faces + Vertices - Edges = 2)